SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.2: Chemical Families — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 2.2: Chemical Families
Driving Question	DRIVING QUESTION: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm — and how do their "family" properties explain the materials that build, light up, and sustain our daily lives?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Phenomenon: Explosions, Glowing Tubes, and Rusty Iron — What Is Going On?**	Students observed three anchor phenomena side by side: (1) a video clip of sodium metal dropped into a water-filled basin — the metal skitters, hisses, and ignites with a violet flame; (2) a photograph of neon and argon discharge tubes glowing in Nairobi street signs; (3) a rusted iron jembe left in a Rift Valley farm field after the long rains. Students recorded initial observations using a See–Think–Wonder chart, noting colour, energy release, and any visible change to the material.	Students learned that the three phenomena involve three very different elements — yet each element's behaviour is not random; it is predictable from the element's identity. Students could not yet explain why, but they established that some elements react violently, some barely react at all, and some react slowly over time. This productive confusion anchors all eight lessons. Pedagogical note: resist the urge to answer student questions at this stage — capture every question on the Driving Question Board (DQB) so learners can revisit them as evidence accumulates.	Students could not yet fully explain the phenomena. However, they generated testable questions such as: 'What makes sodium explode but iron only rust?' and 'Why do some gases glow in tubes without reacting?' These questions seed the DQB. Teacher should cold-call at least 6 students to share one observation each, enforce a 10-second wait time after each prompt, and explicitly validate uncertainty as the starting point of science — referencing how Kenyan chemist and KEBS researcher Dr. Julius Mwangi uses systematic observation before explanation.
Lesson 2 **Sorting the Periodic Table Into Families: Who Belongs Together?**	Students examined a large colour-coded periodic table poster and a set of element property cards (melting point, reactivity with water, electrical conductivity, state at room temperature). Working in groups, they sorted 20 element cards into clusters based on shared properties — mimicking Dmitri Mendeleev's original sorting challenge. Groups then compared their clusters with the accepted periodic table groupings: Group 1 (alkali metals), Group 2 (alkaline	Learners learned that the periodic table is organised into chemical families — alkali metals (Group 1), alkaline earth metals (Group 2), halogens (Group 17), noble gases (Group 18), and transition metals — and that elements within the same group share similar chemical properties because they have the same number of outer-shell (valence) electrons. The periodic table is therefore a predictive tool, not merely a list. Pedagogical note: emphasise that 'group'	Students connected their card-sorting activity to the anchor phenomena: sodium (Group 1) and neon (Group 18) behave differently because they belong to different families with different valence electron counts. Students updated the DQB — moving the question 'What makes elements behave differently?' toward a partial answer: 'Elements in the same column share the same number of outer electrons, which determines their reactivity.' Teacher should

	earth metals), Group 17 (halogens), Group 18 (noble gases), and the transition metals block.	and 'family' are used interchangeably in Kenyan KICD materials; the column number equals the number of valence electrons for Groups 1 and 2.	display a blank group-column diagram and cold-call 4 students to place sodium, magnesium, chlorine, and neon in the correct columns, providing a 15-second wait time.
Lesson 3 **Noble Gases: Why Do Neon and Argon Glow Without Reacting?**	Students observed (via video or live demonstration if available) discharge tubes filled with noble gases — neon (orange-red glow), argon (blue-violet glow), krypton (pale white), and xenon (lavender). They also read a short data table showing that noble gases have full outer electron shells (helium: 2; neon, argon, krypton, xenon: 8 valence electrons) and recorded that their boiling points, ionisation energies, and reactivity values are dramatically different from those of Group 1 or Group 17 elements. Students noted that the Nairobi street-sign phenomenon from Lesson 1 involves argon or neon inside sealed glass tubes excited by electricity — not a chemical reaction.	Students learned that noble gases (Group 18) are chemically inert because their outer electron shells are completely full (the octet rule is satisfied without bonding). They neither gain nor lose electrons under ordinary conditions. The glow in discharge tubes is a physical phenomenon — electrons in the gas are energised by an electric current and release energy as visible light when they return to lower energy levels. Pedagogical note: contrast 'chemical change' (new substance formed) with 'physical change' (no new substance) — the glowing tube is a physical/electronic phenomenon, not a reaction, which surprises most students and deepens conceptual understanding.	Students could now fully explain the second anchor phenomenon: neon and argon glow silently in glass tubes because electricity excites their electrons, but no chemical reaction occurs — they have full outer shells and no tendency to bond. This explanation is added to the DQB as a resolved question. Teacher should ask: 'If noble gases never react, why are they still considered a chemical family?' Cold-call 3 students; expected answer: they share the property of full outer shells and chemical inertness, which is itself a family trait. Connect to Kenyan context: argon is used in welding torches at the Jomo Kenyatta International Airport maintenance hangars to shield molten metal from reacting with oxygen.
Lesson 4 **Alkali Metals in Action: Why Does Reactivity Increase Down Group 1?**	Students watched a carefully sequenced video showing lithium, sodium, and potassium each dropped into water (in a fume hood/controlled lab setting). They recorded observations in a data table: lithium fizzes gently and moves; sodium fizzes vigorously, melts into a ball, and may ignite; potassium ignites immediately with a lilac flame. Students plotted atomic radius and ionisation energy against period number for Group 1 elements and identified the inverse relationship — as atomic radius increases down the group, ionisation energy (energy needed to remove the outer electron) decreases.	Students learned that all Group 1 alkali metals have exactly 1 outer-shell electron; that this outer electron is lost more easily as atomic radius increases down the group because the single valence electron is progressively further from the nucleus and shielded by more inner electron shells. This makes potassium more reactive than sodium, and sodium more reactive than lithium. The reaction with water produces a metal hydroxide and hydrogen gas: e.g., $2Na + 2H_2O \rightarrow 2NaOH + H_2$. Pedagogical note: students frequently confuse 'more electrons = more reactive' — correct this by emphasising that it is the ease of losing the OUTER electron that determines reactivity, not the total electron count.	Students returned to the first anchor phenomenon — sodium exploding in water — and could now explain it using electron configuration and atomic radius. They extended this to predict: 'Caesium (Cs), lower in Group 1, would react even more violently.' Students updated the DQB: 'Why does sodium explode in water?' is now fully answered. Teacher connects to Kenya: sodium compounds (NaOH — caustic soda) are used in soap-making industries in Nakuru and Eldoret; farmers in the Rift Valley use sodium-based compounds as cleaning agents. Cold-call 5 students to write the word equation for potassium reacting with water on the board.
Lesson 5 **Halogens: The Reactive Non-Metals of Group 17**	Students observed images and data for fluorine (pale yellow gas), chlorine (yellow-green gas — used in water treatment at Nairobi's Gigiri water plant), bromine (red-brown liquid), and iodine (purple-grey solid	Students learned that halogens have 7 valence electrons and need only 1 more electron to complete their outer shell, making them highly reactive non-metals that readily form negative ions (halide ions:	Students connected halogen properties to real Kenyan life: chlorine purifies drinking water across Kenyan cities; fluoride (a halide) is added to toothpaste and some Kenyan water supplies to prevent tooth

	that sublimes to a violet vapour). They tested iodine solution on starch (ugali flour paste) and observed the blue-black colour change — a standard qualitative test known to Kenyan biology students from food tests. Students recorded the trend: halogens become less reactive and darker in colour going down Group 17, opposite to the trend in Group 1.	F^-, Cl^-, Br^-, I^-). Reactivity decreases down the group because the incoming electron is added to a shell that is progressively further from the nucleus, making electron gain less energetically favourable. Chlorine is a stronger oxidising agent than bromine, so chlorine can displace bromine from a bromine solution (displacement reaction). Pedagogical note: the iodine-starch test is a powerful cross-strand link to Biology — explicitly make this connection to reinforce that chemistry underpins food science.	decay; iodine is added to cooking salt (iodised salt) to prevent goitre, a public health priority in inland Kenya where iodine-deficient soils are common. Students updated the DQB: 'Why do some elements react violently while others don't?' now has a family-level answer — it depends on the number of valence electrons and the ease of gaining or losing them. Teacher cold-calls 4 students to explain why chlorine is more reactive than iodine using the concept of atomic radius.
Lesson 6 **Magnesium in Action: Group 2 Reactivity, Fertilisers, and the Two-Electron Difference**	Students observed magnesium ribbon burning in air (bright white flame, producing magnesium oxide ash) and magnesium reacting with water (slow at room temperature, faster with steam). They also examined photographs of magnesium deficiency in maize crops in Kisumu County — pale, striped leaves — and compared with healthy crops treated with magnesium-containing fertilisers (dolomite: $CaMg(CO_3)_2$). Students recorded the electron configuration of magnesium (2,8,2) and compared it with sodium (2,8,1) using electron-shell diagrams.	Magnesium (Group 2, electron configuration 2,8,2) has two outer electrons. Both must be lost to form the Mg^{2+} ion. Because the nucleus attracts two outer electrons more strongly than sodium's nucleus attracts one, magnesium is less reactive than sodium — it does not explode in water at room temperature. This illustrates the key principle: Group 2 metals are less reactive than Group 1 metals because removing two electrons requires more energy than removing one. The equations for magnesium reactions are: $Mg + O_2 \rightarrow 2MgO$ (burning); $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$ (with steam). Pedagogical note: use the fertiliser context to anchor abstract electron-loss concepts — students in agricultural counties (Kisumu, Nakuru, Trans Nzoia) will have direct experience with fertiliser application.	Students explained why magnesium reacts with water much more slowly than sodium: it has two outer electrons, both of which must be removed, requiring more ionisation energy. They connected this to the DQB question about the rusty jembe — iron (a transition metal) rusts even more slowly than magnesium reacts with water, hinting that transition metals have their own distinct reactivity pattern. Students also explained the agricultural application: magnesium is essential for chlorophyll production in plants, so magnesium-deficient soils reduce crop yields on Kenyan farms. Teacher should cold-call 3 students to draw and compare electron-shell diagrams for Na, Mg, and Al, noting the trend in outer electron count across Period 3.
Lesson 7 **Transition Metals: The Workers of the Periodic Table — Iron, Copper, and Zinc in Kenyan Life**	Students examined physical samples or photographs of iron (jembe blades, roofing sheets — mabati — used across rural Kenya), copper (electrical wiring in Kenyan homes, copper piping), and zinc (galvanised iron coatings, zinc roofing). They observed and recorded: high melting points, metallic lustre, malleability (can be hammered into sheets), and electrical conductivity. Students also observed iron wool burning in oxygen (rapid, dramatic oxidation) and iron nails placed in salt water over three days (slow rusting — $Fe_2O_3 \cdot xH_2O$). They noted that iron displays	Transition metals form a distinct chemical family with high melting points, hardness, malleability, and good electrical and thermal conductivity. Unlike Group 1 and Group 2 metals, transition metals have partially filled d-subshells, which allow them to form multiple oxidation states (e.g., iron: +2 and +3; copper: +1 and +2), act as catalysts, and form coloured compounds (e.g., copper sulphate is blue; iron(III) oxide — rust — is red-brown). They are far less reactive than alkali metals and do not react with cold water. Pedagogical note: students often ask why iron rusts but copper only tarnishes	Students fully resolved the third anchor phenomenon — the rusty jembe: iron reacts slowly with oxygen and water in the presence of moisture to form hydrated iron(III) oxide (rust). This is much slower than sodium's reaction with water because iron's electrons are held more tightly by a larger nuclear charge and d-orbital electrons. Students connected transition metal properties to Kenyan infrastructure: iron/steel in SGR railway tracks (Standard Gauge Railway, Nairobi to Mombasa), copper in the Olkaria geothermal power plant wiring, zinc galvanising on mabati

	two oxidation states: Fe^{2+} and Fe^{3+}.	green (verdigris) — use this to reinforce that different transition metals have different reactivities, even within the family, because their d-electron configurations differ.	roofing to prevent rust in high-rainfall areas like Kericho. Teacher cold-calls 5 students to give one real-world Kenyan use of a transition metal and justify it using a specific physical or chemical property.
Lesson 8 **Final Explanation: Why Do Chemical Families Determine How Elements Build, Light Up, and Sustain Our Daily Lives?**	Students reviewed all evidence accumulated across Lessons 1–7: the DQB with resolved and partially resolved questions, their cumulative model of the periodic table with annotated family properties, data tables comparing reactivity trends in Groups 1, 2, 17, and 18, and transition metal property charts. They revisited the three anchor phenomena from Lesson 1 (sodium in water, neon discharge tube, rusted jembe) and verified that each phenomenon now has a full, evidence-backed explanation grounded in electron configuration and group position.	Students learned that elements belong to chemical families based on their electron configuration and position in the periodic table. Alkali metals (Group 1, 1 valence electron) are the most reactive metals — reactivity increases down the group. Alkaline earth metals (Group 2, 2 valence electrons) are reactive but less so than Group 1. Halogens (Group 17, 7 valence electrons) are the most reactive non-metals — reactivity decreases down the group. Noble gases (Group 18, full outer shells) are chemically inert. Transition metals have partially filled d-subshells, giving them variable oxidation states, catalytic ability, and structural strength. The number and arrangement of valence electrons is the master variable explaining all family-level behaviour. Pedagogical note: this lesson is the synthesis lesson — do not introduce new content. Focus entirely on students constructing the final explanation in their own words, using evidence from all prior lessons. Assess using the 'CER' framework: Claim, Evidence, Reasoning.	Students delivered full CER (Claim–Evidence–Reasoning) explanations for all three anchor phenomena: (1) Sodium explodes in water because it is a Group 1 alkali metal with 1 loosely held valence electron that is easily lost, releasing hydrogen gas and energy violently. (2) Neon glows in glass tubes because electricity excites its electrons, but it never reacts — its full outer shell (8 electrons) makes it chemically inert. (3) The iron jembe rusts slowly because iron is a transition metal with d-orbital electrons held more tightly; it reacts with oxygen and water over time to form $Fe_2O_3 \cdot xH_2O$. Students also answered the full driving question: chemical families, determined by valence electron configuration, explain which elements are safe to use in electrical wiring (copper — transition metal, conductive, stable), which light up signs (neon/argon — noble gases, inert, excited by electricity), which strengthen fertilisers (magnesium — Group 2, essential for chlorophyll), and which must be protected from rust on Kenyan farms (iron — transition metal, oxidises in moist conditions). Teacher should cold-call 6 students to each explain one family using only the periodic table as a reference — no notes. Final DQB review: confirm all student-generated questions from Lesson 1 have been addressed.